

9.7 Solving Quadratics by Factoring

Lesson Introduction

 Benchmark	MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.		 Learning Target	I can solve a quadratic by factoring.
 Benchmark Coherence	Building On MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.		Working Toward MA.912.AR.3.2 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.	
 Essential Questions	- How can my knowledge of factoring help solve quadratic equations? - When is factoring an appropriate method for solving quadratic equations?			
 Lesson Narrative	In this lesson, students will apply their knowledge of factoring polynomial expressions to solving quadratic equations. Students will be introduced to the zero-product property and learn how their ability to factor can help solve certain quadratic equations.			
 Component Summary	9.7.1 Warm-Up (~5 min.)	Reading a passage about Valerie Thomas and relating satellite dishes to parabolas (SE p. 94)	9.7.3 Guided Instruction (10 min.)	Introduction to solving a quadratic equation by factoring (SE p. 95)
	9.7.2 Activity (15 min.)	Matching equations, graphs, intercepts, and factored expressions Card Sort Activity (SE p. 94)	9.7.4 Try It (10 min.)	Solving quadratic equations by factoring practice (SE p. 97)
	9.7.5 Wrap-Up (~5 min.)	Error Analysis activity on solving a quadratic by factoring. (SE p. 97)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> - Zero product property <u>Additional Terms:</u> - Factoring	Cards for Card Sort Activity	Work through each of these questions prior to teaching. After each question, highlight where students may have misconceptions during the factoring process.	

 Teacher Tips	Common Misconceptions • Students may choose the wrong factors. • Students may use the wrong sign of the factors. • Students may make calculation errors when factoring. • Students may forget to find both solutions after factoring.	Things to Consider • Students have already learned how to factor in earlier lessons. Some students may need additional support for factoring, and not as much on using the zero-product property. Consider your student population when designing your lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder The 9.7.1 Warm-Up for this lesson contains a passage about Valerie Thomas and her role in developing satellite dishes. Students can discuss what they notice and wonder about the relationship between the satellite dish and the parabola.	MA.K12.MTR.4.1: Discussion and Reflection In the 9.7.5 Wrap-Up, students will be provided with two work samples from two different students solving the same question. Students will be asked to determine the error in each work sample and provide an explanation of how to correctly solve the question.
 Differentiate Instruction	9.7.4 Try It provides students with an opportunity for solving quadratic equations by factoring. Students will be able to show mastery of multiple components of the question, allowing for more easily identifiable differentiation opportunities.	
	9.7.2 Activity provides students with an opportunity to engage in a card sort. Consider having students do all three questions with real cards for kinesthetic learners to connect to the content.	
Lesson Notes:		

9.7.1 Warm-Up: Satellite Dishes

Component Overview: Students will read a passage about Valerie Thomas and explore the relationship between satellite dishes and parabolas.



9.7.1 Warm-Up: Satellite Dishes

Valerie Thomas is a scientist and inventor. In 1980, she received a patent for inventing the illusion transmitter. This was an integral piece of technology in developing the digital media formats needed for today's video games, movies, and 3D television.

Thomas worked at NASA from 1964-1995, where she helped develop real-time computer data systems to support satellite operations. She oversaw the creation of the Landsat program there, which supported the first satellite to transmit images from outer space.



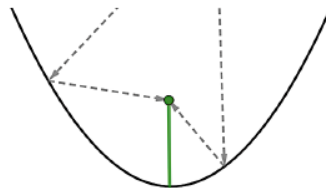
Today, satellite dishes are everywhere. Satellite dishes are designed using a parabolic curve.

1. What key feature of a parabola is the bottom of the dish?

vertex

2. The arrows show how signals bounce off a satellite dish to the receiver. The receiver is always located at a parabola's focus. What key feature does the focus lie on?

Axis of symmetry



Consider a Notice and Wonder learning routine for this Warm-Up. Have students share out everything they notice and wonder as a class. Then release the students to work with a partner or group to answer the remaining questions. Position students to analyze the relationship between the key features of a parabola and the satellite dish. Probe student thinking about other objects in the real-world that may have also been designed using a parabolic curve.



What are other objects in the real-world that are shaped as a parabola?

9.7.2 Activity: Card Sort

Component Overview: Students will engage in a Card Sort Activity to match the equation, graphs, intercepts, and factored expressions. For question 1, students will match the equation cards with the correct card that represents its graph, the card with the correct x -intercept, and the card with the correct y -intercept. Each equation will have three cards that go with it. For question 2, students will factor each expression. For question 3, students will match their expressions from question 2 with the factored expression card from question 1 (P, Q, R, and S) and their x -intercepts. Review the answers to question 3 before moving to question 4. For question 4, have students discuss together and then share their responses.



9.7.2 Activity: Card Sort

1. Work with your partner to match the cards.

Equation	Graph	x -intercept	x -intercept
P	J	D	C
Q	L	A	B
R	K	B	C
S	M	A	D

2. Factor each expression.

Expression T	Expression U	Expression V	Expression W
$2x^2 + 5x - 42$	$2x^2 - 5x - 42$	$2x^2 - 19x + 42$	$2x^2 + 19x + 42$
$(x + 6)(2x - 7)$	$(x - 6)(2x + 7)$	$(x - 6)(2x - 7)$	$(x + 6)(2x + 7)$

3. Work with your partner to match the factored expressions to the x -intercepts.

Expression	Factored Expression	x -intercepts
T	P $(x + 6)(2x - 7)$	$x = -6$ and $x = 3.5$
U	Q $(x - 6)(2x + 7)$	$x = 6$ and $x = -3.5$
V	R $(x - 6)(2x - 7)$	$x = 6$ and $x = 3.5$
W	S $(x + 6)(2x + 7)$	$x = -6$ and $x = -3.5$

4. Discuss with your partner how the factored expression reveals the x -intercepts. Make a conjecture about the connection.

Answers will vary. The x -intercept is the opposite sign of each factor. The x -intercepts are the numbers inside the parenthesis, but with different signs.



Consider having students do all three questions with real cards and then posting them on a large visual like chart paper for kinesthetic and visual learners to connect to the content. In the interest of time, it would be beneficial to have the cards cut out and ready prior to the lesson, to allow more time to be spent on the activity.



Consider making each question an instructional event in the classroom.

- Review the instructions for a question
- Set a time to complete
- Call time
- Select groups to share their response
- Poll class for agree/disagree
- Move to next activity



Try having students switch groups/partners for each question to have different experiences and conversations.



What is the relationship between the factored form and the x -intercepts?

9.7.3 Guided Instruction: Solving Quadratic Equations by Factoring

Component Overview: Students will receive guided instruction on how to solve a quadratic equation by factoring.



9.7.3 Guided Instruction: Solving Quadratic Equations by Factoring

1. In the equation $x \cdot y \cdot z = 0$, which variable has to equal 0 so that the product is equal to 0?

- ☐ x
- ☐ y
- ☐ z
- ☐ *at least one of x , y , or z*

Zero Product Property

If the product of two numbers is 0, then at least one of the numbers is 0.

2. For each expression, use the zero product property to determine the values of x for which the expression would equal 0.

Expression	$(x + 7)(x - 2)$	$(2x + 7)(x - 6)$	$(-3x + 11)(4x + 18)$
x -values	$x = -7$ or $x = 2$	$x = -3.5$ or $x = 6$	$x = \frac{11}{3}$ or $x = -4.5$

Each of these expressions could have been written as quadratic equations that are set equal to 0 such as $x^2 + 5x - 14 = 0$.

3. When solving a quadratic equation set the equation equal to 0, the solution(s) will be the value(s) of x when $y = 0$. Explain how this relates to your conjecture from the card sort.

Answers will vary. This would be similar to finding the x -intercepts during the card sort activity. I set the y equal to zero during the card sort activity.



Make a connection to what students learned about factoring in a previous unit. Highlight key differences, including that in this scenario they are working with equations, not expressions.

9.7.3 Guided Instruction: Solving Quadratic Equations by Factoring (continued)

Another method of solving quadratics is by factoring. When solving by factoring the quadratic must be set equal to 0. Once the quadratic is factored, the zero product property is applied.

4. Solve $h^2 - 10h + 25 = 0$ by factoring.

$$\begin{aligned}(h - 5)(h - 5) &= 0 \\ (h - 5) &= 0 & (h - 5) &= 0 \\ h &= 5 & h &= 5\end{aligned}$$

5. Solve $10f^2 + 13f - 3 = 0$ by factoring.

$$\begin{aligned}(2f + 3)(5f - 1) &= 0 \\ (2f + 3) &= 0 & (5f - 1) &= 0 \\ f &= -\frac{3}{2} & f &= \frac{1}{5}\end{aligned}$$

To solve by factoring, the quadratic equation should be in standard form and set equal to zero.

6. Solve $p^2 + 36 = 13p$ for p by factoring.

$$\begin{aligned}p^2 - 13p + 36 &= 0 \\ (p - 9)(p - 4) &= 0 \\ (p - 9) &= 0 & (p - 4) &= 0 \\ p &= 9 & p &= 4\end{aligned}$$

7. Solve $40x^2 - 18x = 12x + 135$ for x by factoring.

$$\begin{aligned}40x^2 - 30x - 135 &= 0 \\ 5(8x^2 - 6x - 27) &= 0 \\ 5(4x - 9)(2x + 3) &= 0 \\ (4x - 9) &= 0 & (2x + 3) &= 0 \\ x &= \frac{9}{4} & x &= -\frac{3}{2}\end{aligned}$$



Have students relate what they did in their 9.7.2 Activity to what they are experiencing now. Students will begin making the connection between their knowledge of factoring and their ability to solve quadratic equations. Help students build their understanding of the zero-product property by naming the property when setting each factor equal to 0.



Differentiation may be necessary at this stage if students did not do well with the factoring pieces of the 9.7.2 Activity. If students struggle with factoring, this additional step may add too much complexity for the student.



- How do questions 6 and 7 look different than questions 4 and 5?
- What would we need to do before factoring?
- How does this relate to applying the zero-product property?

9.7.4 Try It: Factoring

Component Overview: Students will be given an opportunity to practice solving quadratic equations by factoring. This activity is designed to be done independently. Consider allowing students to check their work with a partner in the activity after trying on their own.



9.7.4 Try It: Factoring

Solve each by factoring. Show your work.

1. $x^2 - 10x - 15 = 9$
 $x^2 - 10x - 24 = 0$
 $(x - 12)(x + 2) = 0$
 $(x - 12) = 0$ $(x + 2) = 0$
 $x = 12$ $x = -2$
2. $-3x^2 + 6x + 22 = 2 - 5x$
 $-3x^2 + 11x + 20 = 0$
 $-(3x^2 - 11x - 20) = 0$
 $-(x - 5)(3x + 4) = 0$
 $(x - 5) = 0$ $(3x + 4) = 0$
 $x = 5$ $x = -\frac{4}{3}$



Consider grouping students together who have a deficit in factoring. Evidence from their performance on the factoring unit and their performance on the 9.7.2 Activity and 9.7.3 Guided Instruction in this lesson should be used to determine which students will need support. Work with these students on the factoring component so that they can complete the work of this concept.



To factor, what must the equation be set equal to? Ask students their strategies for question 2. Students may need to be reminded that the polynomial on the left side of the equation could be moved to the right side of the equation, resulting in a positive leading coefficient.

9.7.5 Wrap-Up: Error Analysis

Component Overview: Students will be provided with sample responses from two students and must analyze their work and correct the students' processes. Allow students to work on this independently so you can assess their level of understanding with the content of this lesson.



9.7.5 Wrap-Up: Error Analysis

The work of two students who solved $2x^2 - 12x + 14 = 2x - 6$ is shown.

Ruchelle	Quinyang
$2x^2 - 12x + 14 = 2x - 6$	$2x^2 - 12x + 14 = 2x - 6$
$2(x^2 - 6x + 7) = 2(x - 3)$	$2x^2 - 12x + 14 - (2x - 6) = 2x - 6 - (2x - 6)$
$\frac{2(x^2 - 6x + 7)}{2} = \frac{2(x - 3)}{2}$	$2x^2 - 14x + 20 = 0$
$x^2 - 6x + 7 = x - 3$	$(2x - 5)(x - 4) = 0$
$x^2 - 5x + 4 = 0$	$x = \frac{5}{2}$ or $x = 4$
$(x - 4)(x - 1) = 0$	
$x = 4$ or $x = 1$	



This activity will provide differentiation data for which students need additional supports in which component of the process. The component the students miss and/or get correct in the error analysis could determine what a student understands to provide differentiation and support if necessary.



For an additional challenge, have students explain or correct the mistake in the process.

- Both Ruchelle and Quinyang made one mistake. Circle each mistake.